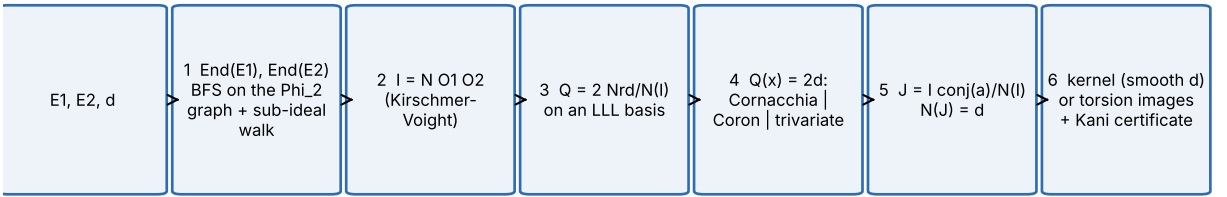
hybrid (smooth d = l^e): guess an l^e1 prefix as sub-ideals after step 1, run 2-5 on the rest, compose before 6



steps 1 and 6 use the Deuring correspondence on E0 over F_(p^4): p < 2^31

steps 2-5 are pure quaternion arithmetic, any p (tested at 61 and 127 bits)